CPE 2300 · RISC-V · TERM PROJECT

Project Workbook

Set your **plan** once. Then each Wednesday has its own page where your two goals appear automatically — record how much you attained, any plan changes, each member's contribution, the core code you wrote, and the challenges you hit. Final deliverables (one code file + report) are due **Friday, December 4 at 11:59 PM.**

WED NOV 11 **Weekly Report** due Wed Nov 11

This week's goals — from your plan; mark how each went

Goal 1: Implement grid initialization and neighbor counting
↳ **Core 1: Implement game rules and logic to simulate successive generations** — Done

Goal 2: Implement and test one complete generation update
↳ **Core 1: Implement game rules and logic to simulate successive generations** — Done

How much of this week's goals were attained?

All

Changes to the plan (if any)

We changed who did what work just because some tasks needed previous tasks to be completed first before starting them, and we had differing schedules. But the work was still evenly split between us all, and the work we finished was what we had originally

Member contributions this week

Will Bakey

Created the clear_grid function to clear all cells to 0, and created the initialize_grid function to set a specific pattern of "live" cells to start the simulation. Initialize_grid also used the load_cell function, which sets the selected cell to the specified state (1 = "alive", 0 = "dead").

Logan Chook

Created the next_generation function, which calculates the next state for each cell from the current grid and loads them into the next grid. Created the swap_buffers function, which takes the current grid and the next grid and swaps the two, effectively moving to

Daiki Hew

Set up the two arrays for the double-buffer grid. Set up .equ values to be used elsewhere in the code. Created the count_neighbors function, which counts how many cells adjacent to the current one are "alive".

Core code developed this week — summarize the progress first, then paste the key routine(s) and link the commit/file

Finished all of the main logic of Conway's Game of Life: setting up two buffers for the two grids, initialization/clearing the grids, neighbor counting, calculating the next generation of cells, and moving into the next generation of cells.

```
# Conway's Game of Life in Assembly
# by Will Bakey, Daiki Hew, and Logan Chook

.data
# -- VGA Display
.equ FB,      0x20000000
.equ PAL,     0x21000000
.equ PRESENT, 0x21001008
# -- Grid/Cell Info (16x12 cell grid)
```

https://github.com/wjbakey/CPE2300_Final_Project/commit/2316b773913ca2cf705ff213474

Challenges faced — what was hard or did not work

Not too many challenges this week, the hardest part was probably getting adjusted to the idea of indexes of a 2D array when the addresses in assembly are 1 dimensional. Through trial and error we figured out how to properly move through indexes using pointers and adding to the them to change where they were pointing.

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